

Agenda

- History
- Web Architecture
- HTML introduction

History of Internet

1. Early Idea (1950s–1960s)

- Scientists began thinking about how computers could communicate.
- J.C.R. Licklider proposed a “Galactic Network” of connected computers.
- The key concept developed: packet switching (sending data in small chunks).

2. ARPANET (1969) – The First Internet

- Created by the U.S. Department of Defense (ARPA).
- Connected 4 universities initially.
- This is considered the birth of the internet.

3. Development of Protocols (1970s)

- TCP/IP (Transmission Control Protocol / Internet Protocol) was created.
- These rules allowed different networks to communicate.
- On January 1, 1983, ARPANET officially adopted TCP/IP → modern internet begins.

4. Expansion (1980s)

- More universities and research institutions joined.
- The Domain Name System (DNS) was introduced (e.g., google.com instead of IP numbers).
- Networks began linking globally.

5. World Wide Web (1990s)

- Invented by Tim Berners-Lee (1989–1991).
- Introduced:
 - Web pages
 - HTML
 - Web browsers
- First popular browser: Mosaic (1993).
- The internet became accessible to the public.

6. Commercialization (Late 1990s–2000s)

- Businesses and websites exploded (Amazon, Google, eBay).
- Dot-com boom (and crash in 2000).
- Rise of email, search engines, and e-commerce.

7. Modern Internet (2010s–Present)

- Social media (Facebook, Twitter, Instagram).
- Smartphones made the internet mobile.
- Cloud computing, streaming, AI, IoT (smart devices).
- Faster networks (4G, 5G).

TCP/IP Protocol

- Handles how data is sent and delivered
- Breaks data into packets and send it
- Ensures:
 - Correct delivery to server or client
 - Order of packets reached are in same sequence
 - No data loss (retransmission if needed)
- IP → Finds the destination (addressing)
- TCP → Ensures reliable delivery
- Works at a lower level (network + transport layer)

Http protocol

- HTTP (HyperText Transfer Protocol) is the set of rules that defines how a browser (client) and a server communicate on the web.
- It is used to
 - Request web pages
 - Send/receive data (HTML, JSON, images)
- **HTTP is a stateless protocol**
- It means each request is independent, Server does NOT remember previous requests

1. HTTP/1.0 (1996)

- First widely used version
- Features:
 - One request per connection
 - Connection closes after each request
 - Slow performance
 - Problem: Every file (HTML, CSS, JS) needed a new connection

2. HTTP/1.1 (1997)

- Major improvement over 1.0
- Features:
 - Persistent connections (keep-alive)
 - Multiple requests on same connection
 - Better caching
 - Host header support
 - Benefit: Faster than HTTP/1.0 and Reduced connection overhead

3. HTTP/2 (2015)

- Modern and high-performance

- Features:
- Multiplexing (multiple requests at once)
- Binary protocol (not text-based)
- Header compression
- Server push (server sends data before request)
- Benefit: Much faster loading and Efficient use of connection

Server

- It is a physical device with highest configuration that is used to process the data on large scale and give out the response
- It generally consists of a-
 1. Web server
 2. DB Server
 3. Languages
 4. Operating System (platform)
- Server can have a Software Stack which is a Collection of softwares
- examples
 1. WISA
 - Windows
 - IIS
 - SQL Server
 - ASP.net
 2. MEAN
 - MongoDB
 - Express
 - Angular
 - Node

Types of Server

1. web server
 - used to serve http or https requests
 - e.g. apache2, nginx, IIS (Internet Information Service), express
2. database server
 - RDBMS: MySQL, MariaDB, Oracle, MS SQL Server, PostgreSQL, IBM DB2 etc.
 - No SQL: MongoDB, redis, firebase, couch db, cocroach db etc.
3. file server
 - used to share files with many users at a same time
 - e.g. ftp (file transfer protocol), nfs (network file system), smb (server message block)
4. mail server
 - used to send and receive emails
 - e.g. Postfix (SMTP server), DoveCot (IMAP and PoP3)

Browser Architecture

- A **web browser is a software application** that allows you to access, locate, and display information, such as web pages, images, and videos, on the World Wide Web
- Components of Browser Architecture are

1. User Interface

- This is the user interface for the browser.
- It includes the Address Bar, back button, Bookmarking options, Refresh button, etc.
- The browser's user interface is not specified in any formal specification, but comes from practices shaped over decades of experience (and browsers copying each other).
- As a result, all browsers have UIs that are extremely similar to each other.

2. The Browser Engine

- The browser engine marshals actions between the browser's user interface and the browser's rendering engine.
- When you type in a new website and press the enter key, the browser UI will tell the browser engine, which will then communicate with the rendering engine.

3. The Rendering Engine

- The rendering engine is **responsible for displaying the requested content**.
- The rendering engine will start by getting the contents of the requested document from the networking layer.
- It takes in the **HTML code and parses it to create the DOM (Document Object Model) tree**.
- Examples of rendering engine include
 1. Safari - Webkit Rendering Engine
 2. **Chrome - Blink Rendering Engine (Blink is a fork of WebKit)**
 3. FireFox - Gecko Rendering Engine

4. Networking Layer

- The Networking Layer is responsible for making network calls to fetch resources.
- It imposes the right connection limits, formats requests, deals with proxies, caching, and much more.

5. JavaScript Engine

- The JavaScript Engine is used to parse and execute JavaScript code on the DOM.
- The JavaScript code is provided by the web server, or it can be provided by the web browser
- Early browsers used JavaScript interpreters, but modern JavaScript engines use **Just-In-Time compilation for improved performance**.
- Examples of JavaScript Engine include
 1. Safari - JavaScriptCore
 2. **Chrome - V8 JavaScript Engine**
 3. FireFox - SpiderMonkey Engine

6. **UI Backend**

- This layer is responsible for drawing the basic widgets like select or input boxes and windows. Underneath it uses operating system UI methods.

- The rendering engine uses the UI backend layer during the layout and painting stages to display the web page on the browser.

7. Data Storage

- The browser needs to save data locally (cookies, cache, etc.) so the Data Storage component handles this part.

Web Architecture

1. Web Server

- A web server is a system (hardware and/or software) that delivers web content, such as websites, web pages, and other resources, to users over the internet or an intranet.
- It handles requests from client devices, typically browsers, and serves the requested resources using standard web protocols like HTTP (Hypertext Transfer Protocol) or HTTPS (HTTP Secure).
- There are many web server software options, each with unique features and use cases. Popular ones include:
 1. Apache HTTP Server: Open-source and widely used, known for flexibility and extensive module support.
 2. Nginx: High-performance and lightweight, often used for handling large amounts of traffic.
 3. Microsoft IIS (Internet Information Services): A web server for Windows environments, integrated with Microsoft technologies.

2. HTTP Request

- An HTTP request is sent by a client to a server to ask for a resource or perform an action.
- It consists mainly of two parts - Header and Body
- Headers Provide additional information about the request.
- Examples:
 - Host: Specifies the target host (e.g., Host: example.com).
 - User-Agent: Identifies the client making the request (e.g., browser or app).
 - Content-Type: Indicates the format of the request body (e.g., application/json).
- Body (Optional): Contains data sent to the server (used in methods like POST or PUT).

3. HTTP Response

- An HTTP response is sent by the server to the client in reply to a request, containing the requested resource, a status code, and other metadata.
- It consists mainly of two parts - Header and Body
- Headers Provide metadata about the response.
- Examples:
 - Content-Type: The format of the response data (e.g., text/html or application/json).
 - Content-Length: The size of the response body in bytes.
 - Set-Cookie: Used to send cookies to the client.
- Body (Optional): Contains the requested resource or additional information.
- Example:
 - HTML for a webpage.
 - JSON data for an API.

URL

- A URL (Uniform Resource Locator) is the address of a unique resource on the internet. It is one of the key mechanisms used by browsers to retrieve published resources, such as HTML pages, CSS documents, images, and so on.
- A URL is composed of different parts, some mandatory and others optional.  alt text

1. Scheme

- The first part of the URL is the scheme, which indicates the protocol that the browser must use to request the resource (a protocol is a set method for exchanging or transferring data around a computer network).
- Usually for websites the protocol is HTTPS or HTTP (its unsecured version)

2. Authority

- Next follows the authority, which is separated from the scheme by the character pattern  /.
- If present the authority includes both the domain (e.g., www.example.com) and the port (80), separated by a colon:
- The domain indicates which Web server is being requested.
- Usually this is a domain name, but an IP address may also be used (but this is rare as it is much less convenient).
- The port indicates the technical "gate" used to access the resources on the web server.
- It is usually omitted if the web server uses the standard ports of the HTTP protocol (80 for HTTP and 443 for HTTPS) to grant access to its resources. Otherwise it is mandatory.

3. Path to resource

- /path/to/myfile.html is the path to the resource on the Web server.
- In the early days of the Web, a path like this represented a physical file location on the Web server.
- Nowadays, it is mostly an abstraction handled by Web servers without any physical reality.

4. Parameters

- ?key1=value1&key2=value2 are extra parameters provided to the Web server.
- Those parameters are a list of key/value pairs separated with the & symbol. - The Web server can use those parameters to do extra stuff before returning the resource.

5. Anchor

- #SomewhereInTheDocument is an anchor to another part of the resource itself.
- An anchor represents a sort of "bookmark" inside the resource, giving the browser the directions to show the content located at that "bookmarked" spot.- On an HTML document, for example, the browser will scroll to the point where the anchor is defined; on a video or audio document

HTTP Request Methods

1. GET: Requests a resource without modifying it.
2. POST: Submits data to the server for processing.

3. PUT: Updates or replaces an existing resource.
4. DELETE: Deletes a resource.

HTTP Response Status Code

1. 1xx (Informational): Request received, continuing process.
 - 100 Continue: Initial part of a request received.
2. 2xx (Success): Request was successful.
 - 200 OK: Request succeeded.
 - 201 Created: Resource was successfully created.
3. 3xx (Redirection): Client needs to take further action.
 - 301 Moved Permanently: Resource has a new URL.
 - 302 Found: Resource temporarily moved.
4. 4xx (Client Errors): Issues with the client's request.
 - 400 Bad Request: The request is malformed.
 - 401 Unauthorized: Authentication required.
 - 404 Not Found: Resource not found.
5. 5xx (Server Errors): Issues on the server side.
 - 500 Internal Server Error: Generic server error.
 - 503 Service Unavailable: Server is overloaded or down.

History of the First Browser and HTML

Tim Berners-Lee's (Inventor of the World Wide Web) Vision

- In March 1989, Tim Berners-Lee wrote a proposal for a system to manage information at CERN, which he called the World Wide Web.
- By the end of 1990, he had developed the code for the first web server, web browser, and editor.
- WorldWideWeb (Nexus) was first web browser
- It served as both a browser and an editor, allowing users to view and create web content.
- The name was later changed to Nexus to differentiate it from the World Wide Web information space.

Browser History

- WorldWideWeb, later named Nexus, was the sole browser for the web until 1992 when Lynx was introduced.
- Lynx, introduced in 1992, was and is a text-based browser. Used primarily for locating, reading, and downloading text-based articles, it continues to be used today for those purposes.
- Mosaic, an early version of Netscape Navigator, was introduced in 1993. It was the first web browser with a graphical user interface, making the web more accessible to the masses.
- Netscape Navigator, a renamed and improved version of Mosaic, was introduced in 1994.
- Microsoft's Internet Explorer 1.0 was introduced in 1995.
- Timeline
 - 1990 → First browser
 - 1995–2001 → IE vs Netscape (1st war)
 - 2004+ → Firefox, Chrome rise (2nd war)
 - Today → Chrome dominance

HTML

- HTML (HyperText Markup Language) is the standard language used to create and structure content on the web.
- It provides the foundation for web pages by defining the structure and elements of the content, such as headings, paragraphs, links, images, tables, and more.
- The latest standard is **HTML5**
- To add html 5 code, start the document with

```
<!DOCTYPE html>
```

- **DOCTYPE**: document type (tag used to start the html document)
- It is **Case in-sensitive**
- To add comment

```
<!-- comment -->
```

Key Features of HTML

1. **Markup Language**: HTML **uses "tags"** to mark up different parts of content, indicating their roles (e.g., heading, paragraph, list).
2. **HyperText**: Enables linking to other web pages or resources using hyperlinks (**<a>** tag).
3. **Platform-Independent**: HTML can be rendered on any device with a web browser.
4. **Extensible**: **Can be combined with other technologies like CSS (for styling) and JavaScript (for interactivity).**

HTML History

1. HTML 1.0 (1993)

- Tim Berners-Lee, the inventor of the World Wide Web.
- Purpose was to share research documents and enable hyperlinking between them.
- Key Features:
 - Basic text formatting (headings, paragraphs, lists).
 - Hyperlinks (**<a>** tag).

2. HTML 2.0 (1995)

- Standardize HTML for broader use.
- Key Features:
 - Support for tables and forms.
 - Basic image embedding (**img** tag).
 - Standardized syntax rules.

3. HTML 3.2 (1997)

- Managed by the World Wide Web Consortium (W3C).
- Introduced features for richer content.
- Key Features:
 - Support for scripting languages like JavaScript.
 - Improved table support for better layouts.
 - Introduction of new elements like for styling.

4. HTML 4.01 (1999)

- W3C continued to refine HTML.
- Enhanced web functionality and accessibility.
- Key Features:
 - Separation of content and style via CSS.
 - Introduction of attributes like id and class for better styling and scripting.
 - Improved support for forms and multimedia.

5. XHTML 1.0 (2000)

- W3C attempted to merge HTML with XML.
- Introduced stricter syntax and better compatibility.
- Key Features:
 - All tags must be properly closed (e.g., `<br />`).
 - Case sensitivity for tags and attributes.
- Challenges:
 - Too strict for practical use; adoption was limited.
 - Developers found it difficult to migrate from HTML 4.

6. HTML5 (2014, Official Recommendation)

- Modernize HTML for web applications and multimedia.
- Key Features:
 - Semantic Elements: `<header>`, `<footer>`, `<article>`, `<nav>` for better content structure.
 - Multimedia Support: Native audio (`<audio>`) and video (`<video>`) elements.
 - Canvas and SVG: For drawing graphics and animations directly in the browser.
 - APIs: Built-in APIs for features like geolocation, drag-and-drop, and offline storage.
 - Backward Compatibility: Maintains support for older HTML versions.
- It provided Simplified coding practices.

Tags

- Word enclosed by `<` and `>` signs
- It is also called as an element
- All tags in HTML are pre-defined by W3C
- E.g. `<h1>`, `<p>`, `<table>`

Types of tags

1. Opening

- Used to open a data/information
- E.g. `<h1>`, `<p>`, `<html>`

2. Closing

- Used to close the data/information
- E.g. `</h1>`, `</p>`, `</html>`

3. Empty

- Tag having no data/content
- Two ways of representing it
 - `<tag></tag>`
 - `<tag />`
 - E.g. `<br/>`, `<hr/>`

4. Root

- Tag which starts and ends the document
- Is also called as Document Type or Document Tag or Document Element
- E.g. `<html>` is root tag for html document